

Andrew (Hyoungjoon) Yoo

70 Morningside Dr, New York, NY 10027 • hy2774@columbia.edu • (207)233-6778

Education

Columbia University

New York, NY

Mathematics, Computer Science. Bachelor of Arts. GPA 3.7

2022-2026

- Relevant Coursework: Probability Theory, Machine Learning, Advanced Algorithms, Analysis of Algorithms, Unsupervised Learning, Artificial Intelligence, Natural Language Processing, Databases, Computational Genomics, Modern Analysis, Modern Algebra, Differential Manifolds, ODE, PDE
- Dean's List, Putnam Top 500

Research Experience

Columbia University Math Department

New York, NY

Math Modeling Research

May 2024 – August 2024

- Researched efficiency optimization of NYC CitiBike network by designing rebalancing schemes to reduce the number of low service stations by 80 percent.
- Applied mathematical modeling and data analysis techniques to real-world research challenges, including dynamical systems, stochastic modeling, simulation optimization, and LP/MIP in python.
- Developed skills in project design, simulation, literature review, and scientific writing.

Directed Reading Program

September 2023 – May 2024

- Researched topics in Algorithms and Measure Theory.
- Solved graduate-level problems collaborating with Math PhD students and presented my findings to peers at the conclusion of each semester.

Markov Mixing Time Reading Group

May 2025 – September 2025

- Participated in a group on Markov Chains and Mixing Times; answered weekly questions and open problems.

Work and Leadership Experience

Northcross Group

Portland, Maine

Data Science Analyst Intern

June 2025 – August 2025

- Built a legal document-processing pipeline that semantically chunked and parsed contracts line-by-line using OCR and transformer-based LLMs, auto-tagging compliance items and boosting review throughput by 400%.
- Delivered a transaction “lookback” tool for banks that flagged posting-order timing errors before settlement and prevented overdraft-fee events via automated checks/alerts.

Putnam Seminar

Seminar Lead

September 2024 – December 2025

- Led weekly meetings for the Columbia Undergraduate Putnam Seminar preparing 30+ students for the Putnam exam.

Columbia Journal of Undergraduate Mathematics

Content Editor

October 2023 – May 2026

- Edited papers in topics in geometry, analysis, and topology for a 100+ page undergraduate math journal with readership of more than 1000.
- Collaborated and communicated with fellow editors to discuss accuracy and accessibility of papers.

Columbia University Computer Science Department

Teaching Assistant

September 2025 – May 2026

- Worked as a teaching assistant for the class Analysis of Algorithms.

Columbia Formula SAE Racing

September 2023 – May 2025

- Designed the carbon fiber suspension in the 2024-2025 Columbia FSAE car using Solidworks.

Reading Team Math

October 2023 – May 2026

- Volunteered in weekly tutoring of elementary school students through an after-school program.

Skills & Interests

Technical: Python(pandas, NumPy, scikit-learn, PyTorch), SQL, Java, MatLab, Solidworks, Git; Methods: regression, time-series, optimization, Monte Carlo simulations

Language: English - Native, Korean - Fluent